Comprehensive Algorithmic Trading Strategy for XAU/USD

A Multi-Timeframe Machine Learning Stacking Approach via OANDA REST v20 Architecture

Author: Verj Labs Inc. Engineering **Asset Class:** Spot Gold (XAU/USD) **Date:** June 2026

1. Executive Summary & Market Philosophy

The global spot gold market (XAU/USD) represents one of the most highly liquid yet structurally volatile trading pairs available to quantitative systems. Traditional high-frequency retail strategies operating exclusively on ultra-short horizons (e.g., the 1-minute $M1$ timeframe) frequently collapse due to high order-execution latency, institutional market-maker manipulation, and random white noise.

This technical blueprint presents a highly robust alternative: a systematic, short-term momentum-scalping framework designed to execute transactions at the close of 5-minute ($M5$) intervals, anchored heavily by macro structural context derived from 30-minute ($M30$) trends. By utilizing a machine learning stacking methodology powered by an optimized **LightGBM (Light Gradient Boosting Machine)** framework, the trading engine maps raw order-book imbalances, execution spreads, and non-linear technical indicators directly into deterministic trading entries.

Rather than attempting to forecast massive macro shifts, this system identifies micro-regimes to target highly probable alpha extractions between 0.05% and 0.20% per transaction, safely neutralizing market noise through strict volatility-adjusted risk mechanics.

2. System Infrastructure & OANDA API Topology

To ensure reliable data delivery and lightning-fast entry execution, the system splits its infrastructure into two asynchronous software systems built directly on top of the **OANDA REST v20 API** ecosystem.

2.1 The Training Loop (Asynchronous Offline Pipeline)

The background training loop operates independently of active trading windows to mitigate local latency. It schedules a weekly fetch of historical data spanning a rolling 5-to-10-year window directly via the HTTP polling endpoint:
`/v20/instruments/{instrument}/candles`. The data acquisition script queries both the $M5$ and $M30$ granularities, storing normalized open, high, low, close, and volume (OHLCV) records inside a local time-series engine. This data forms the master table required for historical feature synthesis and model re-tuning.

2.2 The Inference & Execution Loop (Real-Time Live Engine)

During active trading sessions, the live engine avoids heavy API polling loops by maintaining a persistent WebSocket connection to OANDA's real-time streaming endpoint:
`/v20/pricing/stream`. This streaming topology pushes instant, tick-by-tick updates containing active market

liquidity (*Bid* and *Ask* quotes). Sub-second tick streams are locally aggregated to calculate the instant execution spread and trace intra-candle volume velocity, guaranteeing that the final predictive input table is generated within milliseconds of a candle's closing boundary.

3. Mathematical Feature Engineering Matrix

The LightGBM framework relies on a master tabular dataset constructed precisely at the terminal tick of every closing *M5* candle. The inputs are partitioned into three distinct vector groups to ensure broad structural coverage.

3.1 Group A: Macro Trend Anchors (M30 Granularity)

To protect the scalping system from trading directly against dominant institutional trends, the feature engine calculates macro indicators on the *M30* chart. This forms a structural direction filter:

- **Exponential Moving Averages (EMA_{30}):** Fast (13-period) and slow (34-period) EMAs are tracked to establish multi-hour trend direction and slope.
- **Moving Average Convergence Divergence ($MACD_{30}$):** Calculated via the traditional difference of 12 and 26-period EMAs, paired with a 9-period signal line. The resulting histogram serves as an explicit macro momentum score.

3.2 Group B: Micro Execution Metrics (M5 Granularity)

Immediate trade execution triggers are constructed using technical variables that measure localized price-to-volume relationships on the active *M5* timeframe:

- **Volume Weighted Average Price (*VWAP* Distance):** Because OANDA provides a highly reliable tick volume metric representing transaction frequency, an intraday rolling VWAP is calculated. The distance feature is engineered as:

  D_{vwap} = P_{close} - ext{VWAP}

- **Volume Rate of Change (*VROC*):** Measures localized volume acceleration over a 6-candle rolling window, isolating high-liquidity breakouts from low-volume retail traps.
- **Average True Range (*ATR*):** A standard 14-period True Range moving average used to quantify current volatility.

3.3 Group C: Micro-Structure & Order Book Metrics

This group targets order-flow imbalances and live execution friction, serving as our most predictive real-time inputs:

- **Live Spreads (S_t):** Computed directly from the streaming WebSocket:

$$S_{t} = P_{ask} - P_{bid}$$

- **Order Book Imbalance Ratio ($\mathcal{I}$):** Generated by polling the OANDA `/v20/instruments/XAU_USD/orderBook` snapshot endpoint. The parser calculates the relative depth of pending resting client orders within a tight 20-pip perimeter of the current spot price (P_{spot}):

$$\mathcal{I} = \frac{\sum V_{buy\_limit} \text{ where } (P_{spot} - 2.0) \le P \le P_{spot}}{\sum V_{sell\_limit} \text{ where } P_{spot} \le P \le (P_{spot} + 2.0)}$$

An environment where $\mathcal{I} >> 1.0$ indicates an intensive barrier of resting buy limit orders, functioning as structural support.

> **Architectual Note: Stacking Pipeline Integration**
>
> In production, advanced implementations can append the out-of-sample predictions of secondary deep models (e.g., an LSTM tracking trailing sequences or an ARIMA baseline) directly into this tabular row. This allows the master LightGBM tree to leverage statistical mean-reversion and sequential memory outputs alongside raw market indicators.

4. Double-Target Labeling Methodology

To map real-time data into exact structural actions, training requires the creation of two distinct historical targets based on a look-ahead horizon of **3 future candles (a hard 15-minute window)**.

4.1 LightGBM Classifier Target (Y_{class})

The classification engine uses a customized triple-barrier structure to assign a direction label (0, 1, or 2) to every historical row based on what happens next in the market:

- **Label 2 (Buy Signal):** The future price path breaches an upper threshold set at $+ (2.5 \times S_{t})$ within the 15-minute window, without ever dropping below a trailing stop boundary defined by $- (1.5 \times \text{ATR}_{M5})$.
- **Label 0 (Sell Signal):** The future price path drops below a lower threshold set at $- (2.5 \times S_{t})$ within the 15-minute window, without ever breaching the upper safety barrier defined by $+ (1.5 \times \text{ATR}_{M5})$.
- **Label 1 (Flat/No-Trade Regime):** The price chops sideways within the boundaries for the full 15 minutes, or violently triggers both boundaries within the same window due to unpredicted market whip-saws.

4.2 LightGBM Regressor Target (Y_{reg})

Simultaneously, a regression model is trained on the exact same row matrix. The target value (Y_{reg}) represents the maximum favorable price expansion percentage achieved before the look-ahead window expires or the stop boundary is hit. This provides a clear statistical estimation of the trade's expected magnitude.

5. Real-Time Production Execution Protocol

When the system enters live production, it follows a strict, step-by-step execution protocol at the termination of each $M5$ interval to minimize execution mistakes and protect account capital:

Execution Phase	Procedural Actions & Core Math	Operational Status
1. Context Evaluation	Evaluate $M30$ indicators. If macro trends are down, lock out buy triggers by enforcing `ALLOW_BUYS = False`.	System Interlock
2. Feature Assembly	Compile the real-time row vector using the closed $M5$ candle, live spread, and order-book imbalance snapshot.	Data Core Active
3. Model Inference	Execute dual model evaluation: `Class = Classifier.predict(row)` `Mag = Regressor.predict(row)`	ML Engine Active
4. Cost & Friction Filter	Validate that the predicted magnitude clears transaction costs: *ext{Mag} \ge 2.5 imes S_{live}*. If false, terminate the sequence.	Risk Protection Gate
5. Capital Sizing	Dynamically calculate position volume using a strict 1% account risk limit (see Equation 4).	Capital Allocation
6. API Transmission	Transmit a linked bracket order containing the entry, target profit, and hard stop-loss to OANDA REST endpoints.	Order Finalized

Critical Operational Warning: Dynamic Spread Management

During high-impact global macroeconomic releases (e.g., US Non-Farm Payrolls, CPI announcements, FOMC rate decisions), liquidity providers drastically widen spreads. If the current streaming spread (S_{t}) surges beyond its rolling 20-period moving average, the execution gate triggers a total system override—immediately cancelling live order entries to avoid severe slippage and execution decay.

6. Risk Management & Dynamic Position Sizing

A trading system's long-term profitability relies heavily on its position-sizing framework. The bot relies on a strict capital preservation rule: **no single trade may risk more than 1.0% of the total available account equity.**

Rather than using fixed, arbitrary lot sizes, the lot volume is dynamically calculated for every individual trade entry. The stop-loss distance is mapped to the asset's active volatility using a multiplier of the current 14-period $M5$ Average True Range:

ext{Stop Loss Distance (Pips)} = 1.5 imes ext{ATR}_{M5}

Using the calculated stop-loss distance, the risk budget, and OANDA's specific contract parameters, the engine derives the precise lot size via the following structural formula:

ext{Required Lot Size} = rac{ ext{Account Balance} imes 0.01}{ ext{Stop Loss Distance (Pips)} imes ext{Pip Value Per Standard Lot}}

By using this method, the system automatically downsizes position sizing during erratic, wide-range market conditions, and safely scales up contract sizes during tight, highly predictable consolidation regimes. This ensures uniform dollar exposure across all market types.

7. Failure Protocols & Model Safeguards

To prevent catastrophic market failures, the automated engine features built-in operational guardrails:

1. **Network Latency Interlock:** If an API order transmission experiences round-trip latency exceeding 200ms, the system relies entirely on OANDA's server-side bracket architecture to manage the pre-linked stop-loss and take-profit parameters safely.
2. **Performance Drift Freeze:** The live tracking system continuously compares live trade results against historical training metrics. If the real-time prediction accuracy slips by more than 7.0% within a rolling 50-trade sample, the system executes an emergency pause—disabling automated entries while maintaining data collection until a weekend retraining cycle corrects the drift.